# A Counterexample in Consistency: Why Consistency Does Not Necessarily Imply MSE Convergence

Wesley's Teaching Note in Mathematical Statistics and Econometrics

**Abstract**

This teaching note reviews several fundamental criteria for evaluating point estimators, including unbiasedness, efficiency, consistency, and mean squared error. It then establishes that convergence in mean square implies convergence in probability. The main purpose of the note is to study a counterexample showing that the converse is false: an estimator may converge in probability to the true parameter even though its mean squared error diverges to infinity. The example illustrates an important distinction between the probability of an estimation error and the magnitude of that error. It also demonstrates that consistency alone does not guarantee convergence of expectations, asymptotic unbiasedness, or mean squared error convergence.

## 1 Introduction

Suppose that the objective is to estimate an unknown population parameter $\theta$ using a random sample

$$X_1, X_2, \ldots, X_n.$$

A point estimator of $\theta$ is a statistic of the form

$$T_n = T(X_1, X_2, \ldots, X_n).$$

Because $T_n$ is a function of random observations, it is itself a random variable. Different samples generally produce different estimates. Therefore, the statistical performance of $T_n$ must be evaluated through its sampling distribution.

Several criteria are commonly used to evaluate point estimators:

1. unbiasedness;
2. efficiency;
3. consistency;
4. mean squared error.

These criteria answer different questions. Unbiasedness concerns whether an estimator is correct on average. Efficiency concerns the dispersion of an estimator. Consistency concerns whether the estimator approaches the true parameter as the sample size becomes large. Mean squared error combines both bias and dispersion into a single measure.

The distinctions among these criteria are important. In particular, an estimator can be consistent without being unbiased, and it can even be consistent while having an unbounded mean squared error. The latter possibility is the central subject of this note.

# 2 Basic Review of Point Estimation

## 2.1 Unbiasedness

The bias of an estimator $T_n$ for the parameter $\theta$ is defined as

$$\text{Bias}(T_n) = \mathbb{E}[T_n] - \theta. \tag{1}$$

The estimator $T_n$ is said to be unbiased if

$$\mathbb{E}[T_n] = \theta, \tag{2}$$

or, equivalently,

$$\text{Bias}(T_n) = 0. \tag{3}$$

The expectation here is taken over repeated samples generated from the same population. Therefore, unbiasedness means that the average estimate across repeated samples is equal to the true parameter.

Unbiasedness does not imply that every estimate is close to $\theta$. For example, consider an estimator with the distribution

$$T_n = \begin{cases} \theta - 100, & \text{with probability } \dfrac{1}{2}, \\ \theta + 100, & \text{with probability } \dfrac{1}{2}. \end{cases} \tag{4}$$

Its expectation is

$$\mathbb{E}[T_n] = \frac{1}{2}(\theta - 100) + \frac{1}{2}(\theta + 100) \tag{5}$$

$$= \theta. \tag{6}$$

Thus, $T_n$ is unbiased. Nevertheless, every realization of the estimator is exactly 100 units away from the true parameter. Unbiasedness by itself is therefore not sufficient to guarantee accurate estimation.

## 2.2 Efficiency

Suppose that $T_{1n}$ and $T_{2n}$ are both unbiased estimators of $\theta$. If

$$\mathrm{Var}(T_{1n}) < \mathrm{Var}(T_{2n}), \tag{7}$$

then $T_{1n}$ is said to be more efficient than $T_{2n}$.

Because both estimators are centered at the true parameter, the estimator with the smaller variance has a sampling distribution that is more concentrated around $\theta$.

An unbiased estimator that has the smallest variance among all unbiased estimators is called a minimum-variance unbiased estimator. In large-sample econometrics, researchers also compare the asymptotic variances of consistent estimators. An estimator with a smaller asymptotic variance is regarded as asymptotically more efficient.

Efficiency is therefore a relative concept: it normally compares estimators that satisfy the same unbiasedness or consistency requirement.

## 2.3 Consistency

Consistency is an asymptotic property. An estimator $T_n$ is consistent for $\theta$ if

$$T_n \xrightarrow{p} \theta, \tag{8}$$

where $\xrightarrow{p}$ denotes convergence in probability.

Formally, $T_n \xrightarrow{p} \theta$ means that, for every $\varepsilon > 0$,

$$\lim_{n\to\infty} P\left(|T_n - \theta| > \varepsilon\right) = 0. \tag{9}$$

Equivalently,

$$\lim_{n\to\infty} P\left(|T_n - \theta| \leq \varepsilon\right) = 1. \tag{10}$$

The number $\varepsilon$ represents an arbitrarily small but fixed tolerance. Consistency means that the probability of an estimation error exceeding this tolerance approaches zero as the sample size increases.

It is important to distinguish consistency from unbiasedness:

- Unbiasedness is a finite-sample property concerning $\mathbb{E}[T_n]$ for a given $n$.
- Consistency is a large-sample property concerning the limiting behavior of the entire sampling distribution of $T_n$.

An estimator may be biased in finite samples but still be consistent if its bias and dispersion disappear as $n$ increases.

# 3 An Example of a Biased but Consistent Estimator

Suppose that

$$X_1, \ldots, X_n$$

are independently and identically distributed with

$$\mathbb{E}[X_i] = \mu \qquad \text{and} \qquad \operatorname{Var}(X_i) = \sigma^2 < \infty.$$

The sample mean is

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i. \tag{11}$$

Now define

$$T_n = \left(1 - \frac{1}{n}\right) \overline{X}_n. \tag{12}$$

The expectation of $T_n$ is

$$\mathbb{E}[T_n] = \left(1 - \frac{1}{n}\right) \mathbb{E}[\overline{X}_n] \tag{13}$$

$$= \left(1 - \frac{1}{n}\right) \mu. \tag{14}$$

Its bias is therefore

$$\text{Bias}(T_n) = \mathbb{E}[T_n] - \mu \tag{15}$$

$$= -\frac{\mu}{n}. \tag{16}$$

Unless $\mu = 0$, the estimator is biased for every finite $n$. However,

$$\text{Bias}(T_n) \to 0. \tag{17}$$

By the weak law of large numbers,

$$\overline{X}_n \xrightarrow{p} \mu. \tag{18}$$

Because

$$1 - \frac{1}{n} \to 1, \tag{19}$$

Slutsky's theorem implies that

$$T_n = \left(1 - \frac{1}{n}\right) \overline{X}_n \xrightarrow{p} \mu. \tag{20}$$

Thus, $T_n$ is biased in finite samples but consistent for $\mu$.

This example shows that finite-sample unbiasedness is not necessary for consistency.

# 4 An Unbiased but Inconsistent Estimator

The converse problem can also arise. Suppose again that

$$X_1, \ldots, X_n$$

are independently and identically distributed with

$$\mathbb{E}[X_i] = \mu \qquad \text{and} \qquad \operatorname{Var}(X_i) = \sigma^2 > 0.$$

Consider the estimator

$$T_n = X_1. \tag{21}$$

Although the sample contains $n$ observations, this estimator always uses only the first observation.

It is unbiased because

$$\mathbb{E}[T_n] = \mathbb{E}[X_1] = \mu. \tag{22}$$

However,

$$\operatorname{Var}(T_n) = \operatorname{Var}(X_1) = \sigma^2. \tag{23}$$

The variance does not decrease as the sample size increases because $T_n$ does not make use of the additional observations. Its sampling distribution does not become more concentrated around $\mu$. In general,

$$P(|T_n - \mu| > \varepsilon) = P(|X_1 - \mu| > \varepsilon), \tag{24}$$

which does not approach zero merely because $n$ increases. Thus, $T_n$ is generally not consistent.

This example demonstrates that unbiasedness does not imply consistency.

# 5 Mean Squared Error

The mean squared error of $T_n$ as an estimator of $\theta$ is

$$\mathrm{MSE}(T_n) = \mathbb{E}\left[(T_n - \theta)^2\right]. \tag{25}$$

The MSE measures the expected squared distance between the estimator and the true parameter. Because large estimation errors are squared, the criterion places particularly heavy weight on extreme errors.

## 5.1 Bias–Variance Decomposition

The MSE can be decomposed into a variance component and a squared-bias component:

$$\boxed{\mathrm{MSE}(T_n) = \mathrm{Var}(T_n) + \left[\mathrm{Bias}(T_n)\right]^2}. \tag{26}$$

To derive this result, write

$$T_n - \theta = (T_n - \mathbb{E}[T_n]) + (\mathbb{E}[T_n] - \theta). \tag{27}$$

Squaring both sides gives

$$\begin{aligned}(T_n - \theta)^2 = {} & (T_n - \mathbb{E}[T_n])^2 + (\mathbb{E}[T_n] - \theta)^2 \\ & + 2\,(T_n - \mathbb{E}[T_n])\,(\mathbb{E}[T_n] - \theta).\end{aligned} \tag{28}$$

Taking expectations,

$$\begin{aligned}\mathbb{E}[(T_n - \theta)^2] = {} & \mathbb{E}\left[(T_n - \mathbb{E}[T_n])^2\right] + (\mathbb{E}[T_n] - \theta)^2 \\ & + 2\,(\mathbb{E}[T_n] - \theta)\,\mathbb{E}\left[T_n - \mathbb{E}[T_n]\right].\end{aligned} \tag{29}$$

The final term is zero because

$$\mathbb{E}\left[T_n - \mathbb{E}[T_n]\right] = 0. \tag{30}$$

Therefore,

$$\begin{aligned} \mathrm{MSE}(T_n) &= \mathrm{Var}(T_n) + \left(\mathbb{E}[T_n] - \theta\right)^2 && (31) \\ &= \mathrm{Var}(T_n) + \left[\mathrm{Bias}(T_n)\right]^2. && (32) \end{aligned}$$

This decomposition explains why an estimator with a small bias may have a smaller MSE than an unbiased estimator. A small increase in bias may be desirable if it produces a sufficiently large reduction in variance. This is known as the bias–variance trade-off.

# 6 MSE Convergence Implies Consistency

A useful sufficient condition for consistency is that the MSE approaches zero.

**Theorem 1.** *If*

$$\lim_{n\to\infty} \mathrm{MSE}(T_n) = 0, \tag{33}$$

*then*

$$T_n \xrightarrow{p} \theta. \tag{34}$$

*Proof.* For every $\varepsilon > 0$,

$$|T_n - \theta| > \varepsilon \tag{35}$$

is equivalent to

$$(T_n - \theta)^2 > \varepsilon^2. \tag{36}$$

Because $(T_n - \theta)^2$ is a nonnegative random variable, Markov's inequality gives

$$P(|T_n - \theta| > \varepsilon) = P\left((T_n - \theta)^2 > \varepsilon^2\right) \tag{37}$$

$$\leq \frac{\mathbb{E}[(T_n - \theta)^2]}{\varepsilon^2} \tag{38}$$

$$= \frac{\mathrm{MSE}(T_n)}{\varepsilon^2}. \tag{39}$$

If

$$\mathrm{MSE}(T_n) \to 0, \tag{40}$$

then

$$\frac{\mathrm{MSE}(T_n)}{\varepsilon^2} \to 0. \tag{41}$$

Consequently,

$$P(|T_n - \theta| > \varepsilon) \to 0. \tag{42}$$

By the definition of convergence in probability,

$$T_n \xrightarrow{p} \theta. \tag{43}$$

□

This result may be summarized as

$$\boxed{\mathrm{MSE}(T_n) \to 0 \quad \Longrightarrow \quad T_n \xrightarrow{p} \theta}. \tag{44}$$

Convergence of MSE to zero is also called convergence in mean square, or $L^2$ convergence. Mean-square convergence is stronger than convergence in probability.

The natural question is whether the converse is also true. That is, does

$$T_n \xrightarrow{p} \theta$$

necessarily imply

$$\mathrm{MSE}(T_n) \to 0?$$

The answer is no.

# 7 A Counterexample: Consistency Without MSE Convergence

Let $\theta$ be a fixed and finite parameter. Consider an estimator $T_n$ with the following sampling distribution:

$$T_n = \begin{cases} \theta, & \text{with probability } 1 - \dfrac{1}{n}, \\ n, & \text{with probability } \dfrac{1}{n}. \end{cases} \tag{45}$$

Equivalently, its probability mass function can be written as

$$f_{T_n}(t) = \begin{cases} 1 - \dfrac{1}{n}, & t = \theta, \\ \dfrac{1}{n}, & t = n, \\ 0, & \text{otherwise.} \end{cases} \tag{46}$$

For each $n$, the estimator usually produces the exactly correct value $\theta$. However, with the small probability $1/n$, it produces the extreme value $n$.

The probability of the extreme event approaches zero, but the magnitude of the extreme value increases with $n$. These two forces have different implications for consistency and MSE.

## 7.1 Step 1: Proving Consistency

To establish consistency, consider an arbitrary fixed $\varepsilon > 0$. We need to evaluate

$$P(|T_n - \theta| > \varepsilon). \tag{47}$$

If $T_n = \theta$, then

$$|T_n - \theta| = |\theta - \theta| = 0. \tag{48}$$

Therefore, the event $T_n = \theta$ does not contribute to the probability of an error larger than $\varepsilon$.

If $T_n = n$, then

$$|T_n - \theta| = |n - \theta|. \tag{49}$$

Because $\theta$ and $\varepsilon$ are fixed, there exists a sufficiently large $N$ such that, for every $n > N$,

$$|n - \theta| > \varepsilon. \tag{50}$$

Therefore, for all sufficiently large $n$,

$$P(|T_n - \theta| > \varepsilon) = P(T_n = n) \tag{51}$$

$$= \frac{1}{n}. \tag{52}$$

Taking the limit gives

$$\lim_{n\to\infty} P(|T_n - \theta| > \varepsilon) = \lim_{n\to\infty} \frac{1}{n} = 0. \tag{53}$$

Hence,

$$\boxed{T_n \xrightarrow{p} \theta}. \tag{54}$$

The estimator is therefore consistent. Another way to see the result is to observe that

$$P(T_n = \theta) = 1 - \frac{1}{n} \to 1. \tag{55}$$

Thus, the probability that the estimator is exactly equal to the true parameter approaches one.

## 7.2 Step 2: Calculating the MSE

Using the definition of MSE,

$$\mathrm{MSE}(T_n) = \mathbb{E}[(T_n - \theta)^2]. \tag{56}$$

Because $T_n$ has only two possible values, its MSE can be computed directly:

$$\mathrm{MSE}(T_n) = (\theta - \theta)^2 P(T_n = \theta) + (n - \theta)^2 P(T_n = n) \tag{57}$$

$$= (\theta - \theta)^2 \left(1 - \frac{1}{n}\right) + (n - \theta)^2 \frac{1}{n}. \tag{58}$$

Because

$$(\theta - \theta)^2 = 0, \tag{59}$$

we obtain

$$\mathrm{MSE}(T_n) = \frac{(n - \theta)^2}{n}. \tag{60}$$

Expanding the squared term,

$$\mathrm{MSE}(T_n) = \frac{n^2 - 2n\theta + \theta^2}{n} \tag{61}$$

$$= n - 2\theta + \frac{\theta^2}{n}. \tag{62}$$

Since $\theta$ is fixed,

$$-2\theta \tag{63}$$

is constant and

$$\frac{\theta^2}{n} \to 0. \tag{64}$$

The first term, however, satisfies

$$n \to \infty. \tag{65}$$

Therefore,

$$\boxed{\mathrm{MSE}(T_n) \to \infty}. \tag{66}$$

We have thus constructed an estimator satisfying

$$T_n \xrightarrow{p} \theta \tag{67}$$

while simultaneously satisfying

$$\mathrm{MSE}(T_n) \to \infty. \tag{68}$$

This proves that

$$\boxed{T_n \xrightarrow{p} \theta \quad \not\Longrightarrow \quad \mathrm{MSE}(T_n) \to 0}. \tag{69}$$

# 8 The Intuition: Rare but Catastrophic Errors

At first sight, the result may appear contradictory. The estimator is exactly correct with probability

$$1 - \frac{1}{n} \to 1. \tag{70}$$

How can its MSE nevertheless approach infinity?

The explanation is that consistency and MSE measure different aspects of the sampling distribution.

Consistency asks whether the probability of a non-negligible error approaches zero. In the counterexample, the probability of an error is

$$\frac{1}{n} \to 0. \tag{71}$$

Thus, the estimator is consistent.

MSE also takes account of the magnitude of the error. When the rare error occurs, its squared magnitude is

$$(n-\theta)^2. \tag{72}$$

For large $n$,

$$(n-\theta)^2 \approx n^2. \tag{73}$$

The contribution of the rare event to MSE is therefore approximately

$$\underbrace{\frac{1}{n}}_{\text{probability of error}} \times \underbrace{n^2}_{\text{squared magnitude of error}} = n. \tag{74}$$

Although the probability of the error approaches zero, the squared magnitude of the error increases so rapidly that their product diverges:

$$\frac{1}{n} \times n^2 = n \to \infty. \tag{75}$$

The key lesson is therefore:

> Consistency controls the probability of being far from the true parameter, but it does not by itself control how large the estimation error can be when a rare error occurs.

# 9 A Numerical Illustration

Suppose that

$$\theta = 2 \tag{76}$$

and

$$n = 1000. \tag{77}$$

Then

$$T_{1000} = \begin{cases} 2, & \text{with probability } 0.999, \\ 1000, & \text{with probability } 0.001. \end{cases} \tag{78}$$

The estimator is exactly correct with probability 99.9%. Nevertheless,

$$\begin{aligned} \mathrm{MSE}(T_{1000}) &= (2-2)^2(0.999) + (1000-2)^2(0.001) && (79) \\ &= 998^2(0.001) && (80) \\ &= 996.004. && (81) \end{aligned}$$

Now consider $n = 10{,}000$. Then

$$T_{10{,}000} = \begin{cases} 2, & \text{with probability } 0.9999, \\ 10{,}000, & \text{with probability } 0.0001. \end{cases} \tag{82}$$

The probability of obtaining the correct answer has increased to 99.99%. However,

$$\begin{aligned} \mathrm{MSE}(T_{10{,}000}) &= (10{,}000-2)^2(0.0001) && (83) \\ &= 9998.0004. && (84) \end{aligned}$$

Thus, the estimator becomes more reliable in terms of probability but worse in terms of expected squared loss.

# 10 Bias of the Counterexample

The counterexample also has an unusual bias. Its expectation is

$$\begin{aligned} \mathbb{E}[T_n] &= \theta\left(1 - \frac{1}{n}\right) + n\left(\frac{1}{n}\right) && (85) \\ &= \theta - \frac{\theta}{n} + 1. && (86) \end{aligned}$$

Therefore,

$$\text{Bias}(T_n) = \mathbb{E}[T_n] - \theta \tag{87}$$

$$= 1 - \frac{\theta}{n}. \tag{88}$$

Taking the limit,

$$\lim_{n\to\infty} \text{Bias}(T_n) = 1. \tag{89}$$

Consequently, the estimator is neither unbiased nor asymptotically unbiased.

Recall that asymptotic unbiasedness requires

$$\lim_{n\to\infty} \mathbb{E}[T_n] = \theta, \tag{90}$$

or equivalently,

$$\lim_{n\to\infty} \text{Bias}(T_n) = 0. \tag{91}$$

In the present example,

$$\lim_{n\to\infty} \mathbb{E}[T_n] = \theta + 1 \neq \theta. \tag{92}$$

Nevertheless,

$$T_n \xrightarrow{p} \theta. \tag{93}$$

This shows that convergence in probability does not, by itself, imply convergence of expectations:

$$T_n \xrightarrow{p} \theta \quad \not\Longrightarrow \quad \mathbb{E}[T_n] \to \theta. \tag{94}$$

The reason is again the presence of rare but increasingly extreme realizations.

# 11 Variance of the Counterexample

The variance can be obtained using the bias–variance decomposition:

$$\mathrm{Var}(T_n) = \mathrm{MSE}(T_n) - [\mathrm{Bias}(T_n)]^2. \tag{95}$$

We have already shown that

$$\mathrm{MSE}(T_n) = \frac{(n-\theta)^2}{n} \tag{96}$$

and

$$\mathrm{Bias}(T_n) = 1 - \frac{\theta}{n} = \frac{n-\theta}{n}. \tag{97}$$

Therefore,

$$\mathrm{Var}(T_n) = \frac{(n-\theta)^2}{n} - \left(\frac{n-\theta}{n}\right)^2 \tag{98}$$

$$= (n-\theta)^2 \left(\frac{1}{n} - \frac{1}{n^2}\right) \tag{99}$$

$$= (n-\theta)^2 \frac{n-1}{n^2}. \tag{100}$$

Equivalently,

$$\boxed{\mathrm{Var}(T_n) = \frac{(n-1)(n-\theta)^2}{n^2}}. \tag{101}$$

For large $n$,

$$(n-\theta)^2 \approx n^2 \tag{102}$$

and

$$\frac{n-1}{n^2} \approx \frac{1}{n}. \tag{103}$$

Hence,

$$\mathrm{Var}(T_n) \approx n, \tag{104}$$

so that

$$\mathrm{Var}(T_n) \to \infty. \tag{105}$$

The MSE divergence is primarily driven by the exploding variance generated by the rare extreme realization.

# 12 Why Probability Convergence Does Not Control Expectations

Convergence in probability focuses on probabilities of the form

$$P(|T_n - \theta| > \varepsilon). \tag{106}$$

It does not place a direct restriction on the size of $T_n - \theta$ within the event where the error exceeds $\varepsilon$.

In the counterexample, the probability of the extreme event approaches zero, but the value taken on that event becomes arbitrarily large. This causes expectations involving $T_n$, $(T_n - \theta)^2$, or other unbounded functions of $T_n$ to behave poorly.

Additional conditions are required if one wants to move from convergence in probability to convergence of expectations. One important advanced condition is uniform integrability.

Roughly speaking, uniform integrability prevents a sequence of random variables from placing an economically or statistically important amount of its expectation in increasingly rare and extreme tail events. The sequence in the counterexample fails this requirement because the rare value $n$ becomes increasingly large.

For introductory purposes, the main point is:

> Convergence in probability controls the probability of large deviations, but convergence of expectations also requires sufficient control over the tails of the sampling distributions.

# 13 Summary of the Logical Relationships

The main criteria can be summarized as follows.

| **Property** | **Definition** | **Main Question** |
|---|---|---|
| Unbiasedness | $\mathbb{E}[T_n] = \theta$ | Is the estimator correct on average across repeated samples? |
| Efficiency | Smaller variance among comparable estimators | How concentrated is the sampling distribution around its center? |
| Consistency | $T_n \xrightarrow{p} \theta$ | Does the estimator approach the true parameter as the sample size grows? |
| Asymptotic unbiasedness | $\mathbb{E}[T_n] \to \theta$ | Does the bias disappear as the sample size grows? |
| Mean squared error | $\mathbb{E}[(T_n - \theta)^2]$ | How large is the expected squared estimation error? |

The most important logical relationships are

$$\mathrm{MSE}(T_n) \to 0 \quad \Longrightarrow \quad T_n \xrightarrow{p} \theta, \tag{107}$$

but

$$T_n \xrightarrow{p} \theta \quad \not\Longrightarrow \quad \mathrm{MSE}(T_n) \to 0. \tag{108}$$

Moreover,

$$\text{unbiasedness} \quad \not\Longrightarrow \quad \text{consistency}, \tag{109}$$

and

$$\text{consistency} \quad \not\Longrightarrow \quad \text{finite-sample unbiasedness}. \tag{110}$$

The counterexample studied in this note provides an even stronger result:

$$T_n \xrightarrow{p} \theta \tag{111}$$

can hold even when

$$\text{Bias}(T_n) \to 1, \qquad \text{Var}(T_n) \to \infty, \qquad \text{MSE}(T_n) \to \infty. \tag{112}$$

# 14 Concluding Remarks

Consistency is one of the most important requirements in econometric estimation. It ensures that an estimator becomes arbitrarily close to the true parameter with arbitrarily high probability as the sample size grows. However, consistency is not a complete measure of estimator quality.

The counterexample in this note separates two different dimensions of estimation error. The probability of an error may become negligible, while the magnitude of the error conditional on its occurrence may become enormous. Consistency is determined by the former, whereas MSE depends on both.

For the estimator

$$T_n = \begin{cases} \theta, & \text{with probability } 1 - \dfrac{1}{n}, \\ n, & \text{with probability } \dfrac{1}{n}, \end{cases} \tag{113}$$

the probability of the incorrect outcome is only $1/n$, which approaches zero. This is sufficient for consistency. Nevertheless, the squared error in that rare event is approximately $n^2$. Its expected contribution is therefore approximately

$$\frac{1}{n} \times n^2 = n, \tag{114}$$

which diverges to infinity.

The central lesson can be stated succinctly:

> A vanishing probability of error does not guarantee a vanishing expected loss when the magnitude of rare errors is allowed to grow without bound.

# 15 Supplementary Note: Three Forms of Consistency and Their Logical Relationships

The term "consistency" is sometimes used broadly, but it is important to distinguish among three related convergence concepts:

1. consistency in probability, or weak consistency;
2. almost-sure consistency, or strong consistency;
3. mean-square consistency.

All three concepts describe whether an estimator approaches the true parameter as the sample size increases. However, they impose different requirements on the sequence of estimators.

## 15.1 Weak Consistency

An estimator $T_n$ is weakly consistent for $\theta$ if

$$T_n \xrightarrow{p} \theta. \tag{115}$$

By definition, this means that, for every fixed $\varepsilon > 0$,

$$P(|T_n - \theta| > \varepsilon) \to 0. \tag{116}$$

Weak consistency concerns the probability of observing an estimation error larger than a fixed tolerance. As $n$ increases, this probability must approach zero.

When an econometrics textbook simply states that an estimator is "consistent," it usually means consistency in probability unless another mode of convergence is explicitly specified.

## 15.2 Strong Consistency

An estimator $T_n$ is strongly consistent for $\theta$ if

$$T_n \xrightarrow{a.s.} \theta, \tag{117}$$

where $\xrightarrow{a.s.}$ denotes almost-sure convergence.

Formally,

$$P\left(\lim_{n\to\infty} T_n = \theta\right) = 1. \tag{118}$$

An equivalent formulation is that, for every $\varepsilon > 0$,

$$P\left(\text{there exists an } N \text{ such that } |T_n - \theta| \leq \varepsilon \text{ for every } n \geq N\right) = 1. \tag{119}$$

The phrase "almost surely" means that the convergence occurs for every sample path except possibly for a set of sample paths having probability zero.

This is a pathwise statement. For almost every infinite sequence of observations, the estimator eventually enters any arbitrarily small neighborhood of $\theta$ and remains there.

## 15.3 The Difference Between Weak and Strong Consistency

Weak consistency and strong consistency consider estimation errors from different perspectives.

Weak consistency examines the probability of an error separately at each sample size:

$$P(|T_n - \theta| > \varepsilon) \to 0. \tag{120}$$

It says that the probability of an error becomes small as $n$ increases. However, it does not necessarily rule out the possibility that errors continue to occur infinitely often along some sample paths.

Strong consistency examines the entire infinite sample path. It requires that, with probability one, the estimator eventually remains close to the true parameter.

The distinction can be summarized informally as follows:

> Weak consistency means that the probability of making an error at sample size $n$ approaches zero. Strong consistency means that, along almost every sample path, the estimator eventually stabilizes around the true parameter.

## 15.4 Almost-Sure Convergence Implies Convergence in Probability

Almost-sure convergence is stronger than convergence in probability.

**Theorem 2.** *If*

$$T_n \xrightarrow{a.s.} \theta, \tag{121}$$

*then*

$$T_n \xrightarrow{p} \theta. \tag{122}$$

To understand this implication, define the event

$$A_n(\varepsilon) = \{|T_n - \theta| > \varepsilon\}. \tag{123}$$

If $T_n \to \theta$ almost surely, then for almost every sample path, the event $A_n(\varepsilon)$ can occur only finitely many times. In other words,

$$P\left(A_n(\varepsilon) \text{ infinitely often}\right) = 0. \tag{124}$$

This pathwise convergence implies that

$$P(A_n(\varepsilon)) = P(|T_n - \theta| > \varepsilon) \to 0. \tag{125}$$

Therefore,

$$T_n \xrightarrow{p} \theta. \tag{126}$$

Thus, strong consistency is a sufficient condition for weak consistency.

## 15.5 Mean-Square Consistency

An estimator $T_n$ is mean-square consistent for $\theta$ if

$$\mathbb{E}[(T_n - \theta)^2] \to 0. \tag{127}$$

Because

$$\mathrm{MSE}(T_n) = \mathbb{E}[(T_n - \theta)^2], \tag{128}$$

mean-square consistency is equivalent to

$$\mathrm{MSE}(T_n) \to 0. \tag{129}$$

As established in the preceding section, Markov's inequality gives

$$P(|T_n - \theta| > \varepsilon) \leq \frac{\mathbb{E}[(T_n - \theta)^2]}{\varepsilon^2}. \tag{130}$$

Consequently,

$$\mathrm{MSE}(T_n) \to 0 \quad \Longrightarrow \quad T_n \xrightarrow{p} \theta. \tag{131}$$

Mean-square consistency is therefore another sufficient condition for weak consistency.

## 15.6 Two Sufficient Routes to Weak Consistency

Two commonly used routes for establishing weak consistency are therefore

$$\boxed{\begin{aligned} T_n \xrightarrow{a.s.} \theta \quad &\Longrightarrow \quad T_n \xrightarrow{p} \theta, \\ \mathrm{MSE}(T_n) \to 0 \quad &\Longrightarrow \quad T_n \xrightarrow{p} \theta. \end{aligned}} \tag{132}$$

The first route establishes almost-sure convergence, while the second establishes convergence in mean square. Either result is sufficient to conclude that $T_n$ is consistent in probability.

These are two important sufficient routes, but they are not the only methods for proving consistency. For example, many econometric consistency proofs apply a law of large numbers and the continuous mapping theorem directly.

## 15.7 Why the Implications Cannot Generally Be Reversed

Although both almost-sure convergence and mean-square convergence imply convergence in probability, convergence in probability does not generally imply either of the stronger

results.

In general,

$$T_n \xrightarrow{p} \theta \quad \not\Longrightarrow \quad T_n \xrightarrow{a.s.} \theta, \tag{133}$$

and

$$T_n \xrightarrow{p} \theta \quad \not\Longrightarrow \quad \mathrm{MSE}(T_n) \to 0. \tag{134}$$

The counterexample developed in the next section establishes the second failure. It constructs an estimator that converges in probability to $\theta$, even though its MSE diverges to infinity.

The first failure arises because small error probabilities at each individual sample size do not necessarily prevent errors from occurring infinitely often along a particular sample path.

## 15.8 Example: Convergence in Probability Without Almost-Sure Convergence

Let $A_1, A_2, \ldots$ be a sequence of independent events satisfying

$$P(A_n) = \frac{1}{n}, \tag{135}$$

and define

$$T_n = \mathbf{1}_{A_n}, \tag{136}$$

where $\mathbf{1}_{A_n}$ is an indicator variable equal to one when $A_n$ occurs and zero otherwise.

For every $0 < \varepsilon < 1$,

$$P(|T_n - 0| > \varepsilon) = P(T_n = 1) \tag{137}$$

$$= P(A_n) \tag{138}$$

$$= \frac{1}{n} \to 0. \tag{139}$$

Therefore,

$$T_n \xrightarrow{p} 0. \tag{140}$$

However,

$$\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty. \tag{141}$$

Because the events are independent, the second Borel–Cantelli lemma implies

$$P(A_n \text{ infinitely often}) = 1. \tag{142}$$

Thus, with probability one, $T_n = 1$ occurs infinitely many times. The sequence does not eventually remain close to zero, and hence

$$T_n \not\xrightarrow{a.s.} 0. \tag{143}$$

This example illustrates why convergence in probability cannot generally be used to infer almost-sure convergence.

# 16 Neither Strong Consistency nor Mean-Square Consistency Implies the Other

Almost-sure convergence and mean-square convergence are both stronger than convergence in probability. Nevertheless, they impose different kinds of restrictions and cannot generally be ordered relative to each other.

Almost-sure convergence imposes a pathwise requirement. Mean-square convergence controls the expected squared magnitude of estimation errors. Without additional assumptions, neither one implies the other.

## 16.1 Mean-Square Convergence Does Not Necessarily Imply Almost-Sure Convergence

Return to the sequence

$$T_n = \mathbf{1}_{A_n}, \tag{144}$$

where the events $A_n$ are independent and

$$P(A_n) = \frac{1}{n}. \tag{145}$$

Because $T_n$ is an indicator variable,

$$T_n^2 = T_n. \tag{146}$$

Therefore,

$$\begin{aligned} \mathbb{E}[(T_n - 0)^2] &= \mathbb{E}[T_n^2] && (147)\\ &= P(A_n) && (148)\\ &= \frac{1}{n} \to 0. && (149) \end{aligned}$$

Thus,

$$T_n \xrightarrow{L^2} 0. \tag{150}$$

However, as shown above, the events $A_n$ occur infinitely often with probability one, so

$$T_n \not\xrightarrow{a.s.} 0. \tag{151}$$

Hence,

$$T_n \xrightarrow{L^2} 0 \quad \not\Longrightarrow \quad T_n \xrightarrow{a.s.} 0. \tag{152}$$

## 16.2 Almost-Sure Convergence Does Not Necessarily Imply Mean-Square Convergence

Let

$$U \sim \text{Uniform}(0,1) \tag{153}$$

and define

$$T_n = n\,\mathbf{1}_{\{U \le 1/n\}}. \tag{154}$$

For every realization $U > 0$, there exists a sufficiently large $N$ such that

$$\frac{1}{n} < U \qquad \text{for every } n \ge N. \tag{155}$$

Therefore,

$$T_n = 0 \qquad \text{for every } n \ge N. \tag{156}$$

Since $P(U = 0) = 0$, it follows that

$$T_n \xrightarrow{a.s.} 0. \tag{157}$$

However, its mean squared error is

$$\begin{aligned}
\mathbb{E}[(T_n - 0)^2] &= \mathbb{E}\left[n^2 \mathbf{1}_{\{U \le 1/n\}}\right] && (158)\\
&= n^2 P\left(U \le \frac{1}{n}\right) && (159)\\
&= n^2 \left(\frac{1}{n}\right) && (160)\\
&= n. && (161)
\end{aligned}$$

Consequently,

$$\mathbb{E}[(T_n - 0)^2] \to \infty. \tag{162}$$

Thus,

$$T_n \xrightarrow{a.s.} 0 \quad \not\Longrightarrow \quad T_n \xrightarrow{L^2} 0. \tag{163}$$

This example shows that even pathwise convergence does not guarantee good behavior under squared-error loss. Rare but extremely large values may have a negligible effect on almost-sure convergence while dominating the MSE.

## 16.3 Summary of the Convergence Hierarchy

The principal implications can be summarized as

$$\boxed{\begin{aligned} T_n \xrightarrow{a.s.} \theta &\Longrightarrow T_n \xrightarrow{p} \theta, \\ T_n \xrightarrow{L^2} \theta &\Longrightarrow T_n \xrightarrow{p} \theta. \end{aligned}} \tag{164}$$

The reverse implications do not generally hold:

$$\boxed{\begin{aligned} T_n \xrightarrow{p} \theta &\not\Longrightarrow T_n \xrightarrow{a.s.} \theta, \\ T_n \xrightarrow{p} \theta &\not\Longrightarrow T_n \xrightarrow{L^2} \theta. \end{aligned}} \tag{165}$$

Moreover, without additional regularity conditions,

$$\boxed{T_n \xrightarrow{a.s.} \theta \quad \text{and} \quad T_n \xrightarrow{L^2} \theta \quad \text{do not generally imply each other.}} \tag{166}$$

The reason is that almost-sure convergence and mean-square convergence control different features of the estimator:

- Almost-sure convergence controls the long-run behavior of almost every sample path.
- Mean-square convergence controls the expected squared magnitude of the estimation error.
- Convergence in probability controls only the probability that the estimation error exceeds a fixed tolerance.

Consequently, strong consistency and mean-square consistency should be viewed as two different sufficient routes to weak consistency, rather than as two steps in a single universal hierarchy.